

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Pharm. Examination
University Theory Examination November/December 2015
PH404 Pharmaceutical Analysis - III

Date: 01.12.2015, Tuesday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q. 1 Attempt any **TWO** of the following

- A State Beer's law. Discuss the factors that cause deviation from beer's law. 10
- B Discuss the radiation sources and detectors used in UV – Visible Spectrophotometer. 10
- C (1) Comment on following statements: 06
- a) Quartz is preferred for sample cell construction for UV spectroscopy.
 - b) Methanol does not show absorbance in UV region.
 - c) PDA detector is preferred over other detectors in UV spectroscopy study.
- (2) Explain following terms: 04
- a) Chromophore
 - b) Auxochrome
 - c) Bathochromic shift
 - d) Specific Absorbance

Q 2 Attempt any **TWO** of the following

- A (1) What is Luminescence Spectrometry? and Discuss theory of fluorescence and phosphorescence. 05
- (2) Explain the following factors affecting fluorescence intensity 05
- a) Structure
 - b) Dissolved oxygen
 - c) Solvent
- B (1) Draw a labeled diagram of FT – IR and discuss its principle. 05
- (2) Enumerate the radiation source of infra red spectroscopy and explain in detail any one. 05
- C (1) Write a short note on ATR spectroscopy. 05

- (2) Name the probable functional groups responsible for the following peaks in IR spectra 05
- a) 1716 cm^{-1}
 - b) 2964 cm^{-1}
 - c) broad band at 3339 cm^{-1}
 - d) doublet at 2820 and 2720 cm^{-1}
 - e) 3010 cm^{-1}

SECTION- II

Q 3 Attempt any **TWO** of the following

- A** (1) Differentiate TLC and HPTLC 05
- (2) Enlist basic components and steps involved in HPTLC technique. 05
Write brief note on sample application step for HPTLC analysis.
- B** Write ideal requirements for detectors used in GC. Enlist the GC detectors. 10
Explain working principle, advantages and disadvantages of ANY TWO GC detectors.
- C** Write a detailed note on HPLC detectors 10

Q 4 Attempt any **TWO** of the following

- A** 1) What is Gas Chromatography? Classify. 02
- 2) Explain the principle of Gas Chromatography. 02
- 3) Draw block diagram of GC Instrumentation. 02
- 4) Classify different types of columns used in GC. 02
- 5) What are the requirements of analyte for GC analysis? Give examples 02
of commonly used carrier gases in GC analysis.
- B** What is HETP? Write van Deemter equation. Discuss the equation with 10
reference to factors affecting band broadening in HPLC.
- C** Write note on ANY TWO of the following 10
- 1) Ion Exchange Chromatography
 - 2) Gel Filtration Chromatography
 - 3) Affinity Chromatography